A Comparative Analysis of Two Capacitor-Less LDO Topologies in 130nm CMOS Technology

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***Abstract*—This paper presents the design and comparison of two capacitor-less (CL) PMOS low dropout (LDO) regulators implemented in a 130nm CMOS technology. Each LDO topology was designed using a different OTA (5T OTA, Telescopic OTA). Both LDO configurations drive a 50 pF load, deliver a load current of $500\,\mu\mathrm{A}$, support programmable output voltages ($2.3\,\mathrm{V}$, $2.1\,\mathrm{V}$ and $1.6\,\mathrm{V}$), and have a consumption strictly below $100\,\mu\mathrm{A}$. Simulation results demonstrate that the 5T OTA provides the most balanced final results, whereas the Telescopic OTA maximizes loop gain at the severe expense of the Error Amplifier's output voltage swing.**

I. Introduction

Low dropout (LDO) voltage regulators are one of the most essential building blocks in power-management systems. Especially with today's increasing demand for system-on-chip solutions, many mixed-signal systems utilize LDO regulators to deliver local supply voltages with reduced noise.

This work will focus on designing two individual LDO architectures. Both LDO topologies must meet the proposed specifications that are presented in Table I. The general LDO schematic is also shown in Figure 1.

TABLE I
Technical LDO Target Specifications

Parameter	Target Value
Power Supply Voltage (V_{EXT})	3 V
Bias Current (I_{bias})	5 µA
Load Current (I_L)	500 µA
Load Capacitance (C_L)	50 pF
Reference Voltage (V_{REF})	1.2 V ± 1 %
Output Voltages (V_{OUT})	2.3 V, 2.1 V, 1.6 V
Compensation Capacitor (C_c)	< 7 pF
Phase Margin @ No-Load	> 60°
Phase Margin @ Full-Load	> 70°
Loop Gain @ 1 Hz	> 50 dB
PSRR @ 1 Hz	> 65 dB
PSRR @ 1 MHz	> 2 dB
PSRR @ 10 MHz	> 2 dB
Output-Referred Noise @ 10 Hz	$< -115\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$
Output-Referred Noise @ 1 kHz	$< -125\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$
Output-Referred Noise @ 10 kHz	$< -134\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$
Output-Referred Noise @ 100 kHz	$< -143\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$
Output-Referred Noise @ 1 MHz	$< -155\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$

Section II provides the theoretical background necessary to understand LDO architecture. Sections III and IV present the two topologies designed and tested in this work, while Section V compares their performance.

II. Background

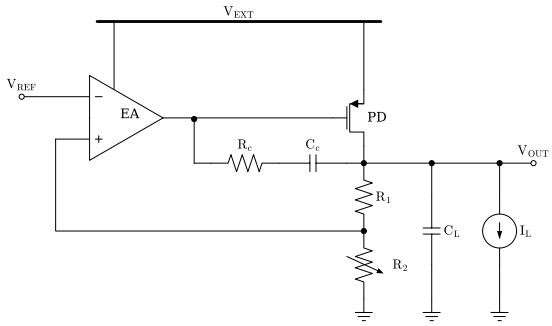


Fig. 1. General LDO Schematic

As shown in Figure 1, a PMOS LDO is made of an Error Amplifier (EA), a PMOS Pass Device (PD), two feedback resistors R_1 and R_2 for setting the output voltage and a compensation network consisting of a nulling resistor R_c and a capacitor C_c. In this work only one-stage EAs will be presented and analyzed.

A. Output Voltage

Assuming that the EA has infinite DC gain (therefore $A_{EA,0} \to \infty$), LDO's gain is given by:

$$\frac{V_{OUT}}{V_{REF}} = \frac{A_{EA,0}}{1 + A_{EA,0}\beta} \xrightarrow{A_{EA,0}\to\infty} \frac{V_{OUT}}{V_{REF}} = \frac{1}{\beta} \tag{1}$$

where β is the feedback factor set by $1 + \frac{R_1}{R_2}$. Therefore:

$$\frac{V_{OUT}}{V_{REF}} = \frac{R_1}{R_1 + R_2} \to \frac{R_1}{R_2} = \frac{V_{REF}}{V_{OUT}} - 1 \tag{2}$$

B. Stability

A small signal model of the LDO schematic can be seen in Figure 2 where the EA's transconductance, output resistance and output load are represented by $g_{m,EA}$, R_1 and C_1 respectively. Similarly, $g_{m,PD}$, R_2 and C_2 represent the same values for the PD. The LDO's transfer function has two poles (given that we have a one stage EA), ω_{p1} (dominant) and ω_{p2} (non-dominant) and a zero ω_z, all of which can be approximated with:

$$\omega_{p1} = \frac{1}{g_{m,PD} R_1 R_2 C_c} \tag{3}$$

$$\omega_{p2} = \frac{g_{m,PD}}{C_1 + C_2} \tag{4}$$

$$\omega_z = \frac{1}{C_c(1/g_{m,PD} - R_c)} \tag{5}$$

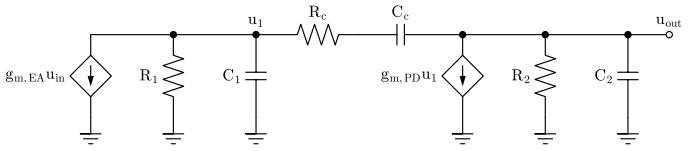


Fig. 2. Small Signal Model of LDO

C. Power Supply Rejection Ratio (PSRR)

Power supply rejection refers to the amount of voltage ripple at the output of the LDO coming from the input voltage and therefore is of great importance. The PSR transfer function of a LDO regulator strongly depends on the type of the EA and whether the input transistors of the EA are NMOS or PMOS (Type-A or Type-B EA respectively). For a PMOS PD, Type-A EAs yield the best PSRR results at low frequencies [2]. Therefore, both LDOs presented in this work utilize Type-A EAs and their PSRR can be approximated by:

$$|PSRR_{DC}| \cong \frac{1}{\beta A_{EA,0} g_{m,PD} r_{ds,p}} \tag{6}$$

At higher frequencies, the nulling resistor along with the sizing of the PD can help in achieving the target PSRR requirements.

D. Noise

Noise in LDO regulators refers to the flicker and thermal noise generated by transistors and resistors. Aside from that, the reference voltage can also contribute to the total noise in LDO. If $S_{n,ref}(f)$, $S_{n,EA}(f)$, $S_{n,PD}(f)$, $S_{n,R1}(f)$ and $S_{n,R2}(f)$ represent the noise power spectral density of the voltage reference, error amplifier, pass device and feedback resistors respectively, then the total output noise power spectral density of the LDO regulator is [2]:

$$\begin{aligned} S_{n,o}(f) \cong (S_{n,ref}(f) + S_{n,EA}(f)) \left(1 + \frac{R_1}{R_2}\right)^2 \\ + S_{n,R2}(f)(\frac{R_1}{R_2})^2 + S_{n,R_1}(f) \end{aligned} \tag{7}$$

The EA and feedback resistors noise are typically the dominant sources of a LDO regulator noise.

III. 5T OTA-Based LDO

The 5-transistor OTA used as EA in this LDO variation can be seen in Figure 3. 5T OTA is a simple differential pair with a gain of:

$$A_{5T} = \frac{g_{m,12}}{g_{5T,out}} \tag{8}$$

whereas the output impedance:

$$g_{5T,out} = g_{ds,2} + g_{ds,4} \tag{9}$$

as for the output swing, in order to keep all the transistors in saturation, we must have:

$$V_{REF} - V_T \leq V_{OUT} \leq V_{EXT} - V_{DSAT} \tag{10}$$

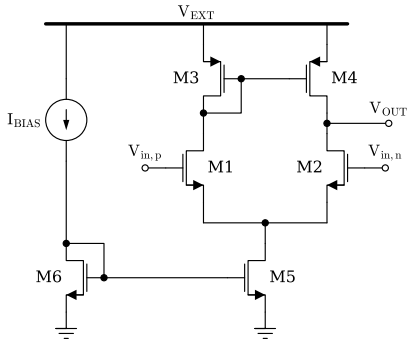


Fig. 3. 5T OTA

TABLE II
Device Sizing for the 5T OTA LDO

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	$42.0/5.0$	µm
M_3, M_4 (Active Load)	$3.8/5.0$	µm
M_5 (Tail Current)	$12.0/5.0$	µm
M_6 (Input Current)	$5.0/5.0$	µm
M_{PD} (Pass Device)	$20.0/0.14$	µm
R_1 (Divider)	15.00	kΩ
R_{21} (Divider)	16.36	kΩ
R_{22} (Divider)	20.00	kΩ
R_{23} (Divider)	45.00	kΩ
R_c (Compensation)	15.00	kΩ
C_c (Compensation)	7.00	pF

In order to satisfy both the PSRR and output voltages, the EA has to deliver enough gain. This can be achieved by increasing the transconductance of the NMOS input pair and the output resistance. Large transistors will also reduce the Flicker noise (as we see later) leading to reduced output-referred noise at lower frequencies. Table II shows the device sizing for this LDO variation.

To verify the fundamental operation of the topology, the DC regulation capabilities are evaluated first. Figure 4 illustrates the LDO's ramp response where we can see that the system settles at the target voltages of 2.3 V, 2.1 V, and 1.6 V with a minimal steady-state offset. This precise tracking confirms that the error amplifier possesses sufficient DC loop gain to minimize steady-state error. Both the loop gain and the loop phase are presented in Figure 6 which also shows the system's phase margin under load. By changing the value of the nulling resistor and therefore the system's zero, the phase margin can either be decreased or increased.

Figure 6 demonstrates the PSRR of the proposed LDO. Simulation results indicate that the width of the PMOS active load directly affects the PSRR performance; consequently, a small PMOS active load was selected. The nulling resistor and the pass device sizing further enhance PSRR at higher frequencies.

Regarding output-referred noise, larger input transistors reduce flicker noise at lower frequencies, as previously mentioned. At higher frequencies, increasing the input transistor current reduces thermal noise contributions.

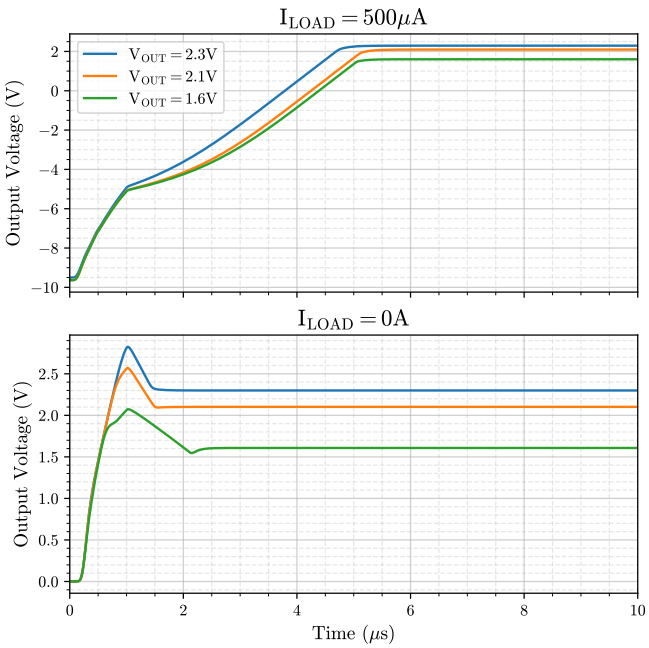


Fig. 4. 5T OTA-Based LDO's Ramp Response

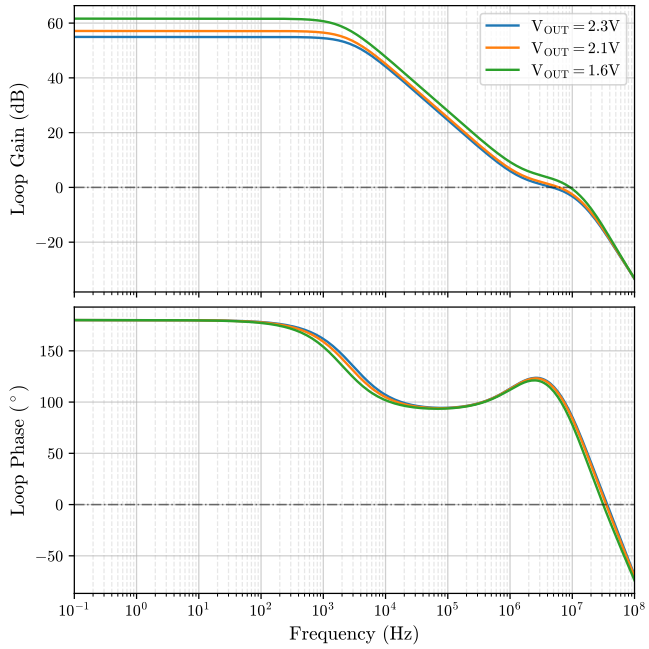


Fig. 5. 5T OTA-Based LDO's Bode Plot Under Load

IV. Telescopic OTA-Based LDO

Figure 8 shows the Telescopic OTA schematic used in this LDO variation. This topology features a cascode gain stage with a high-swing current mirror and self-bias network. Consequently, both loop gain and output impedance are substantially increased compared to the 5T OTA. More specifically, the loop

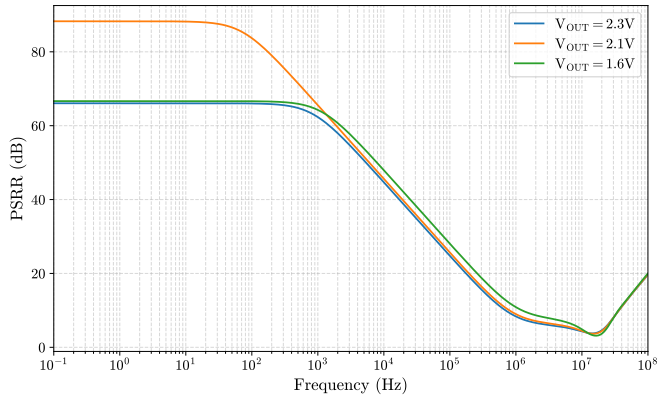


Fig. 6. 5T OTA-Based LDO's PSRR

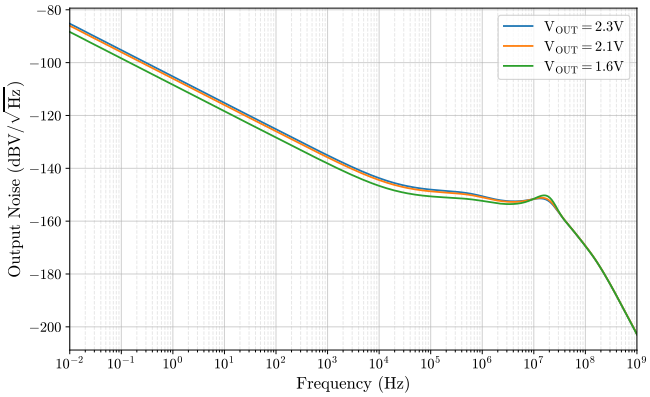


Fig. 7. 5T OTA-Based LDO's Output Referred Noise

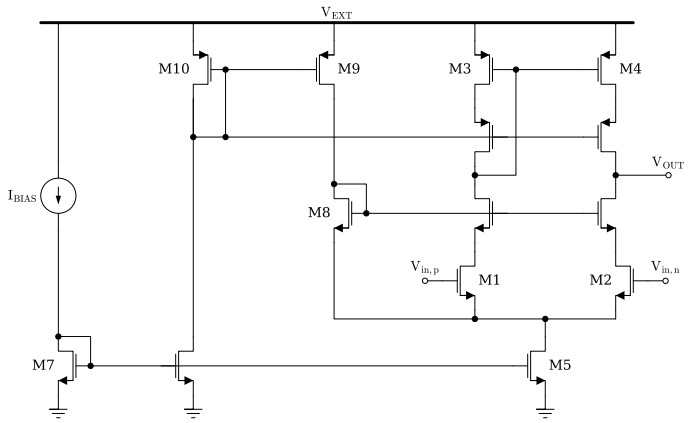


Fig. 8. Telescopic OTA

gain can be given by:

$$A_{TEL} = \frac{g_{m,12}}{g_{tel,out}} \tag{11}$$

whereas the output impedance is given by:

$$g_{tel,out} = g_{ds,2}\frac{g_{ds,2C}}{g_{m,2C}} + g_{ds,4}\frac{g_{ds,4C}}{g_{m,4C}} \tag{12}$$

Note that M2C and M4C represent NMOS and PMOS cascode transistors, respectively.

Due to the stacked cascode transistors, the output swing is much more limited in this OTA. In order to ensure that the OTA has the maximum output swing, the bias transistors should be sized using the following equations:

$$\frac{W_8}{L_8} = \frac{W_{3,4}}{4L_{3,4}} \tag{13}$$

$$\frac{W_{10}}{L_{10}} = \frac{W_{1,2}}{4L_{1,2}} \tag{14}$$

Leading to the following output swing:

$$V_{REF} - V_T + V_{DSAT} \leq V_{OUT} \leq V_{EXT} - 2V_{DSAT} \tag{15}$$

Table III presents the device sizing. Compared to the 5T OTA design, the Telescopic OTA employs smaller bias transistor dimensions, larger feedback resistors, and increased pass device width to ensure the pass device gate voltage remains within the OTA output range.

Figure 9 demonstrates that all three target output voltages are achieved. Figure 10 presents the Bode plot for this LDO variation, which exhibits significantly higher DC gain compared to the 5T OTA, resulting in superior PSRR performance.

TABLE III
DEVICE SIZING FOR THE TELESCOPIC OTA LDO

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	25.0/5.0	µm
M_{1C}, M_{2C} (NMOS Cascode)	0.65/1.0	µm
M_{3C}, M_{4C} (PMOS Cascode)	2.0/1.0	µm
M_3, M_4 (Active Load)	40.0/5.0	µm
M_8 (NMOS Bias)	0.65/4.0	µm
M_9, M_{10} (PMOS Bias)	2.0/4.0	µm
M_5 (Tail Current)	0.95/1.0	µm
M_6 (Bias Current)	0.28/1.0	µm
M_7 (Input Current)	0.28/1.0	µm
M_{PD} (Pass Device)	40.0/0.13	µm
R_1 (Divider)	22.00	kΩ
R_{21} (Divider)	24.00	kΩ
R_{22} (Divider)	29.33	kΩ
R_{23} (Divider)	66.00	kΩ
R_c (Compensation)	20.00	kΩ
C_c (Compensation)	7.00	pF

Figure 11 illustrates the PSRR results, which are superior to those of the 5T OTA due to the higher loop gain. Figure 12 presents the output-referred noise, which is comparable to the 5T OTA results because the cascode transistors contribute negligibly to the total noise.

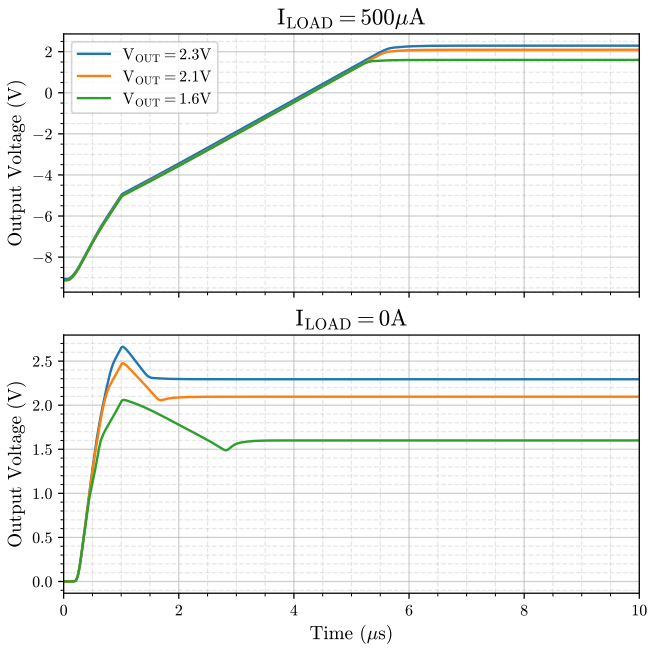


Fig. 9. Telescopic OTA-Based LDO's Ramp Response

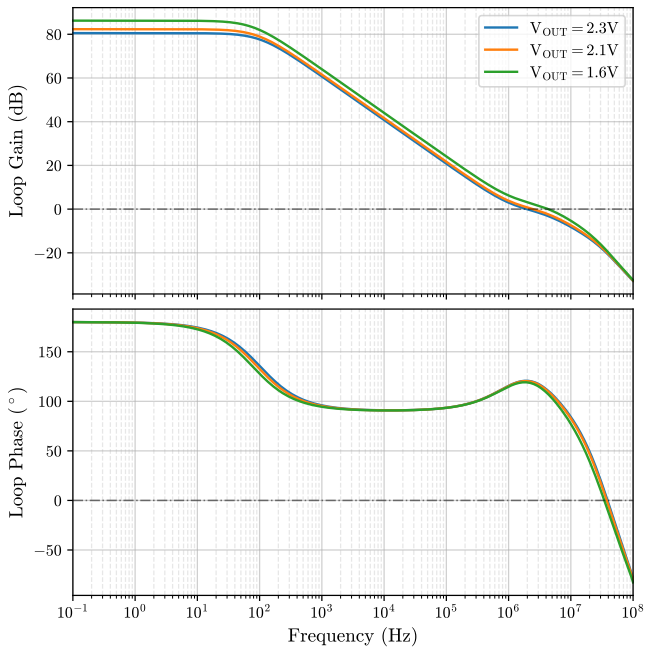


Fig. 10. Telescopic OTA-Based LDO's Bode Plot Under Load

V. COMPARISON

Table IV summarizes the performance of both LDO variations against the target specifications. Both designs satisfy all specifications except the output-referred noise requirement at 1 MHz. The 5T OTA design is preferred when a smaller pass device or increased output voltage swing is required. The Telescopic OTA, conversely, maximizes loop gain and PSRR but suffers from severely limited output swing.

VI. CONCLUSION

Both designs performed well across almost all proposed specifications and failed only to meet the output-referred noise specification at higher frequencies. From the results obtained, it is safe to say that a one-stage EA, such as the 5T OTA or Telescopic OTA, is more than sufficient for building high-performance LDO circuits. Achieving the noise specification at higher frequencies requires increasing the input-transistor bias

TABLE IV
SIMULATED PERFORMANCE SUMMARY (ACROSS ALL TRIMS)

Parameter	Target Specification	5T OTA (Min – Max)	Telescopic-OTA (Min – Max)
Trim Voltage 1	$= 2.3\,\mathrm{V}$	2.293 – 2.299	2.294
Trim Voltage 2	$= 2.1\,\mathrm{V}$	2.096 – 2.102	2.096
Trim Voltage 3	$= 1.6\,\mathrm{V}$	1.602 – 1.607	1.6
Quiescent Current	$< 100\,\mu\mathrm{A}$	48.16 – 94.49	52.56 – 84.12
Phase Margin ($I_{load} = 0$)	$> 60°$	77.62 – 99.2	74.15 – 99
Phase Margin ($I_{load} = 500\,\mu\mathrm{A}$)	$> 70°$	82.7 – 116.2	107.2 – 120.1
Loop Gain @ 1 Hz	$> 50\,\mathrm{dB}$	54.95 – 61.63	80.53 – 86.23
PSRR @ 1 Hz	$> 65\,\mathrm{dB}$	66.07 – 88.27	70.73 – 75.86
PSRR @ 1 MHz	$> 2\,\mathrm{dB}$	8.43 – 10.95	6.373 – 8.537
PSRR @ 10 MHz	$> 2\,\mathrm{dB}$	4.353 – 4.934	2.107 – 2.118
Output-Referred Noise @ 10 Hz	$< -115\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$	-118.3 – -115.1	-118.6 – -115.5
Output-Referred Noise @ 1 kHz	$< -125\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$	-138.1 – -135	-138.1 – -135.1
Output-Referred Noise @ 10 kHz	$< -134\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$	-146.5 – -143.6	-145.4 – -142.5
Output-Referred Noise @ 100 kHz	$< -143\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$	-150.6 – -148	-148 – -145.3
Output-Referred Noise @ 1 MHz	$< -155\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$	-152.3 – -150.5	-149.9 – -148.3

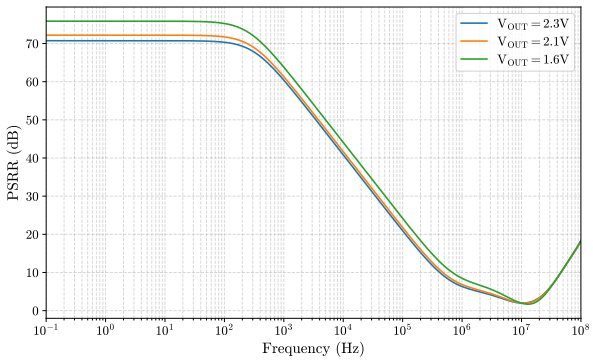


Fig. 11. Telescopic OTA-Based LDO's PSRR

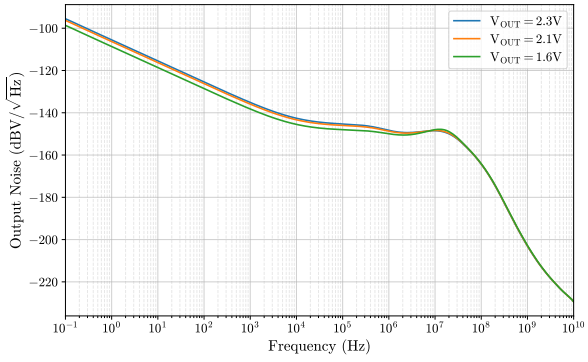


Fig. 12. Telescopic OTA-Based LDO's Output Referred Noise

current, as thermal noise dominates in this regime. However, increasing the bias current in both topologies reduces the output impedance, thereby decreasing loop gain and PSRR performance. Alternative architectures with separate input and gain stages, each optimized for different operating currents, warrant investigation. One such candidate is the current-mirror OTA topology shown in Figure 13, which potentially offers independent optimization of noise and gain performance.

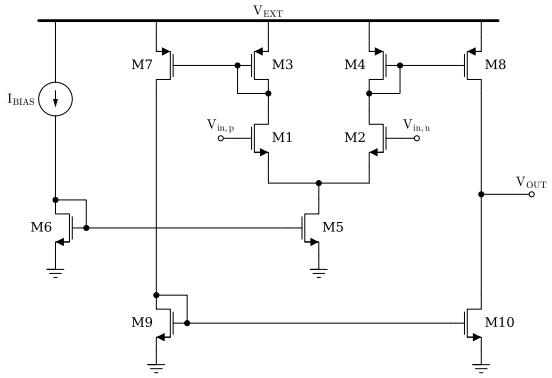


Fig. 13. Current Mirror OTA

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